

Mathan Kumar P

Aspiring Software Engineer | DSA • Backend Systems • AI/ML
+91-8870981422 | mathankumarpalani2006@gmail.com | [LinkedIn](#) | [Github](#)

EDUCATION

SRM Institute of Science and Technology, Kattankulathur
B. Tech in Computer Science and Engineering | CGPA: 9.29 /10

2024 – 2028 (Expected)

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript — OOP proficiency across all

Web & Frameworks: React, Next.js, Node.js, NestJS, Flask, FastAPI, Express.js Flutter, HTML/CSS, Tailwind CSS

AI / ML: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, LLMs, RAG, Agentic AI, Generative AI, NLP (spaCy, BioBERT)

Databases & DevOps: PostgreSQL, MySQL, MongoDB, Firebase, Supabase | Git, Docker, AWS, GCP, Linux, Postman

CS Fundamentals: Data Structures & Algorithms, Operating Systems, DBMS, Object-Oriented Programming, System Design

PROJECTS

Infinite Helix – AI Medical Report Intelligence Platform | Team Lead

FastAPI · spaCy · BioBERT · PostgreSQL · OCR

- Architected a full-stack medical analytics platform leveraging **OCR, NLP (spaCy/BioBERT), and ML** to extract, interpret, and risk-score clinical lab reports across **30+ health parameters**.
- Designed FastAPI microservice backend with cloud PostgreSQL; incorporated observability and reliability best practices to support structured storage and retrieval at scale.

Nichely AI – Niche Discovery & Market Intelligence Platform | Team Lead

React · FastAPI · Python · NLP · Machine Learning

Built an AI-driven market intelligence platform leveraging NLP and ML to analyze trends, keywords, and user intent, delivering real-time niche recommendations for strategic decision-making.

- Implemented optimized inference pipelines and interactive dashboards for processing large-scale unstructured data with high relevance scoring across concurrent user sessions.

ShieldGuard AI – Real-Time Phishing Detection Extension | Individual

JavaScript · DOM Analysis · Cybersecurity · NLP

- Built a browser extension detecting phishing in real time by analyzing DOM structure and URL patterns, flagging malicious sites before user interaction.
- Engineered lightweight client-side inference with NLP-based pattern recognition to identify social engineering cues, delivering instant threat assessment with minimal page overhead.

GPU vs CPU Deep Learning Benchmark Dashboard | Team

PyTorch · CUDA · NVML · Streamlit

- Benchmarked CNN training on CPU vs GPU (CUDA), capturing **training time, throughput, and power consumption** via NVML; built interactive Streamlit dashboard for real-time comparison.

CERTIFICATIONS

- AWS AI/ML Badge – Amazon Web Services** – Demonstrated foundational knowledge in cloud-based machine learning services.
- Google Cloud Platform – 7 Skill Badges** including Cloud Compute, Cloud Storage, Generative AI, LLMs, and Responsible AI.
- NPTEL Certifications** – Object-Oriented Programming & Java | **Coursera** – Generative AI: Introduction and Applications
- Cybersecurity Workshop (Level 1)** – Received **Certificate of Excellence (Best Student)** for Ethical Hacking module.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Solved 150+ DSA problems** across LeetCode (100+), Code360 (50+), strong in arrays, trees, graphs, and dynamic programming.
- Excellence Award** – Advanced DSA Certification by IITM Pravartak & Coding Ninjas; recognized for outstanding performance in advanced DSA training.
- Top 10 – Synaptic Frontier, IIT Madras** – recognized national-level AI/ML hackathon among high-caliber teams.
- Runner-Up – Hack4Health Hackathon** – secured 2nd place and won ₹5,000 cash prize for a healthcare-domain solution.
- Finalist – YUVA Ai-Thon** – Built a creative, impactful solution in a competitive team setting.

EXPERIENCE

Software Engineering Intern– Diances

6 Weeks

- Engineered an AI/ML-powered recommendation system using collaborative filtering; collaborated with cross-functional teammates to gather user requirements and deliver a production-ready backend service, applying best practices across the full development lifecycle — design, implementation, testing, and deployment.

Research Intern – SRMIST

Ongoing

- Conducting applied AI/ML research under faculty mentorship, applying engineering principles to solve complex problems and iterating based on peer feedback.

Consulting project - Vistalane Dynamic Private limited (via SRMIST)

Ongoing

- Contributing to a real-world industry project involving system design and implementation

LEADERSHIP & EXTRACURRICULARS

- Rashtrapati Awardee & Rajapusar Awardee** — Recognized at the national level for outstanding leadership, discipline, and community contribution
- Organizer — Ultron 8.0 Hackathon (SRMIST)** — Led end-to-end organization of a large-scale hackathon, managing team coordination, logistics, participant onboarding, and event operations.